

The rationality of explanation or human capacity? Understanding the impact of explainable artificial intelligence on human-AI trust and decision performance

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ABSTRACT

Artificial intelligence models can process massive amounts of data and surpass human experts in predictions. However, the lack of trust in algorithms sealed in the “black box” is one of the most challenging barriers to taking advantage of AI in human decision-making. Improving algorithm transparency by presenting explanations is one of the most common approaches to curing this. Explainable artificial intelligence (XAI) has been a recent research focus, but most concentrate on explainable algorithm development rather than human factors. Thus, the objective of this study is twofold: (1) to explore whether or not XAI can improve human performance and trust in AI in the competitive tasks of sales prediction, and (2) to reveal the different impact routines XAI on individuals with different task-related capacities. Based on a quasi-experimental study, our results indicate that XAI can improve human decision accuracy in the scenario of sales prediction in cross-border e-commerce. XAI cannot improve self-report trust to AI but can improve behavioral trust. We also found the *placebo* effect of explanation for relatively low task-related capacity.

1. Introduction

With the remarkable technological advances in artificial intelligence (AI), we are reaping the benefits of the advanced algorithm and learning model. AI has been increasingly used in decision-making in various fields, including managerial and organizational contexts (Ivanov & Webster, 2024), citizen services in public sectors (Vassilakopoulou et al., 2023), medical and health care (Ben-jamens et al., 2020), commercial settings such as sales forecasting field (Paschen et al., 2020), etc. It is evidenced that, with AI assistance in decision-making, especially in complex tasks, humans can achieve superior performance than separately by using the complementary strengths of human intelligence and AI.

Though AI can perform impressively in processing a vast amount of data and outperform human experts in making predictions, human decision-makers may be skeptical about adopting AI predictions. The accuracy of AI models depends on the quality of the input training data set, which may suffer from biases or unknown flaws. The results of AI predictions on a specific condition cannot be guaranteed (Zhang et al., 2020). Additionally, the algorithms and statistical models behind AI are complex and considered a “black box,” making it challenging for human decision-makers to assess whether the AI predictions are reasonable (Lucic et al., 2020). The lack of trust in AI is one of the most challenging barriers to taking full advantage of AI in decision-making (Dellermann et al., 2019). The technological approach to help decision-makers know whether to adopt AI predictions is to improve algorithm transparency.

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Explaining the algorithm, which is usually sealed in a “black box” is believed to improve human trust in AI. Explainable artificial intelligence (XAI) has been a recent research focus. Still, most concentration from tech companies is on explainable algorithm development rather than human factors (Babic et al., 2021).

Despite offering explainable AI in human decision-making, the impact of explanation on human-AI trust-building and human-AI joint decision-making performance remains controversial. Some studies found that with the immaturity of existing AI and the widespread skepticism in society about the AI application (Hengstler et al., 2016), the trust toward AI tends to be low (Burton et al., 2020), and offering explanations for AI cannot improve (Zhang et al., 2020) even negatively impact trust in AI (Schmidt et al., 2020). Controversy, some studies found that when making numeric estimates, people would feel more comfortable with algorithm advice compared with human advice (Logg et al., 2019), offering explanations can significantly increase people’s trust in AI (Biran & Cotton, 2017; Wang et al., 2016a).

The controversial findings on the effect of explanation on human trust in AI during AI-assisted decision-making might be twofold: the difference in the decision task scenario and the human factor of task-related capacity. Firstly, prior evidence of the impact of explanation on human-AI interaction is mainly from non-commercial experiment tasks, such as text classification of deception detection (Lai & Tan, 2019), income prediction (Zhang et al., 2020), where the decision making is isolated among participants and non-competitive. Trust building is risk-sensitive. People would behave differently in trust-building at different risk levels (Mcknight et al., 2002), especially in the commercial scenario where the decision is competitive. Thus, in our study of competitive commercial decision-making tasks, the first research question is:

RQ1: Does explanation improve AI-assisted human decision performance and human trust in AI in competitive commercial decision-making?

Additionally, with inconsistent findings on the impact of XAI on human trust in AI, it is worthy of further exploration that decision-makers with different task-related capacities may have other trust-building routines during human-AI interaction. The presence of XAI is with two attributes, the cue of the visualized explanation and the content quality of the explanation. The elaboration likelihood model stated that people could be persuaded via two routines regarding their elaboration likelihood level, the peripheral routine via information cue and the core routine via information content quality (Petty & Cacioppo, 1986). However, prior studies didn’t distinguish the different impact routines of explanation on human trust in AI, i.e. the impact as a transparency cue and the effect via the explanation content quality, respectively. People with different task-related capacities may vary in their interaction with AI, and the way they build trust differs accordingly. Distinguishing the impact of the content quality and cue of explainable AI could help understand the dual trust-building routines in human-AI interaction. Therefore, it is necessary to investigate this question by incorporating the human factor of task-related capacity into the human-AI interaction. Therefore, the second research question for this study is:

RQ2: How does XAI’s impact on people differ according to their task-related capacity?

2. Related work and hypotheses

2.1. Trust in IT artifact

Human interaction with new technology is one of the leading research focuses in information systems and management scholars. Prior researches primarily focus on users’ acceptance and use of new technology based on the model by Davis (1989). Recently, trust in technology has been added as the main predictor of technology acceptance and use (Lee & See, 2004). The concept of trust is initially used to describe human-human or human-organization relationships in the social psychology discipline and is now mitigated in the human-technology relationship (Glikson & Woolley, 2020; Hoff & Bashir, 2015). Trust in technology allows us to unravel the misuse of new technology, i.e., inappropriate over-reliance or under-reliance on technology, and thus has attracted increasing attention among scholars.

There are various definitions of trust from different disciplines, including interpersonal, organizational, and human-automation interaction. Mayer et al. (1995) defined the widely accepted concept of trust as ability, benevolence, and integrity. Later, Mcknight et al. (2002) described trust in e-commerce and empirically measured it as trusting beliefs dealing with benevolence, competence, and integrity. Based on the trust definition by Mayer et al. (1995), lots of scholars explored trust in online environments such as e-commerce. In the mitigation of the concentration of interpersonal trust to human-technology trust, (Nass et al., 1993) proposed the “computers-as-a-social actor” (CSA) paradigm, denoting that the trust perception of computers is similar to the trust of social people. However, some scholars argue that interpersonal trust cannot serve as a sound theoretical base for trust in the IT artifacts (Söllner et al., 2012). The grounds of trust, such as “benevolence,” can only be applied to humans, while computers are impossible to “actively decide whether to keep the interest of the trustor in mind or not” (Lankton et al., 2015; Völter et al., 2021). Thus, (Lee & See, 2004) summarized the basis of trustee characteristics that impact trust as performance, process, and purpose in human-automation interaction. For instance, the foundation of trust by Mayer et al. (1995) can be summarized as performance (ability), process (integrity), and purpose (benevolence). The performance-process-purpose trust framework has been widely used to explore human-automation trusts, such as trust in the AI chatbots (Aoki, 2020). Among the exploration, trust is measured as a self-report or behavioral trust. Among IS scholars, most trust measurements are based on self-report survey data following the integrated measurement model proposed by Gefen et al. (2003). However, self-report data collected by surveys was found to be an unreliable measurement of trust in AI-assisted decision-making fields (Schaffer et al., 2019). Thus scholars tried to use behavioral indicators to measure trust, such as switch percentage or agreement percentage related to AI’s prediction (Zhang et al., 2020), agreement and switch fraction when conducting the tasks when provided with AI’s prediction (Yin et al., 2019), etc. Thus, this study will measure trust with self-report questions and

behavioral trust to investigate whether discrepancies exist between two trust measurements in competitive decision-making tasks.

2.2. Explainable AI in decision-making performance

AI decision aids are infused within decision-making tasks and collaborate with humans to achieve better performance than the isolated operation, conceptualized as “hybrid intelligence” (Dellermann et al., 2019). AI can complement human intelligence in logic and computation speed, but the AI prediction models are opaque and difficult to understand. Thus, explanations of how an AI model works are needed to improve algorithm transparency. The EU general data protection regulation (GDPR) stated the data subject should have the right to “obtain an explanation of the decision reached”.¹ Two types of algorithms with different ways of dealing with opacity exist: interpretable AI and explainable AI (Hancox-Li, 2020). Interpretable models use transparent ‘white-box’ functions with a comparatively easy mathematical and logical structure that can be explained directly from the structure, such as linear models. While, explainable AI is used for post-hoc explanations for ‘black-box’ models, which are too complex to provide a simple explanation. The explanation for “black-box” models is a second algorithm that is used to fit the “black-box” predictions (Babic et al., 2021).

Certain studies illustrated the impact of explainable AI on human decision performance in different task settings. AI-assisted performance is the human performance when provided with the AI prediction (Hemmer et al., 2021). While different in experimental tasks, prior experiments have different findings. Some found that with an explanation of the AI prediction results, human decision-making performance would become better than that of human decision alone, for instance, Choudhary et al., 2021; Lai & Tan, 2019; Wang et al., 2016a. Hemmer et al. (2021) reviewed 72 experiments that measured AI-assisted and XAI-assisted performance and found 46 of them evidenced that XAI-assisted performance outperformed that of AI-assisted alone.

Some results present a non-significant or even negative impact on human prediction accuracies. For instance, Schmidt et al. (2020)’s study found a negative impact of explanation on human prediction accuracy and their trust in AI in the task of text classification. Zhang et al. (2020)’s experiment on the annual income prediction task found the explanation didn’t show significant improvement in the AI-assisted prediction. A similar non-significant impact of XAI on human decision-making of explainable AI was found by Alufaisan et al. (2020), indicating that providing AI prediction is enough to improve user decision accuracy regardless of the presence of explanation or not.

However, extant experimental evidence on the impact of explanation on AI-assisted decision-making performance was primarily based on low-stake and uncompetitive decision tasks. Researchers found that individuals may trust algorithmic advice more regarding accuracy problems such as predictions or investment decisions (Logg et al., 2019). Thus, the type of decision task might impact the influence of explanation on human decision performance. In the current study, we set the decision-making to be competitive commercial tasks, and participants are informed that they should outperform their peers to win the corresponding level of monetary reward. Therefore, it is reasonable to hypothesize that explainable AI improves human performance in the current study scenario of competitive commercial decision-making. Consistent with the measurement of performance in previous studies such as Hemmer et al. (2021) and Liu et al. (2021), we measure human performance in terms of decision accuracy, and decision time. Thus, we propose that in competitive commercial prediction tasks, *explainable AI improves human decision accuracy (H1a) and decreases human decision time (H1b)*. We recruited a control group to complete all prediction trials with an AI assistant, and the experiment group to make their predictions with an explainable AI assistant (refer to Section 3.4 for details). Participants’ decision accuracy and average time cost for each prediction task will be compared to test H1a and H1b.

2.3. Explanations and trust in AI

To grasp the advantages of artificial intelligence, trust in AI is necessary for effective knowledge transfer between machine algorithms and human intelligence (Ostheimer et al., 2021). Providing transparency for machine intelligence is considered an effective approach to trust-building. Explanations can help to learn and are shown to improve positive user perceptions of intelligent systems. Thus, it is believed that explanations can improve the performance achieved with the aid of intelligent systems (Gregor & Benbasat, 1999).

The impact of transparency on human trust in autonomous agents and algorithms has been widely investigated in multiple disciplines, including information systems, computer science, and psychology. With the merit of making the system performance transparent, explanations can improve user acceptance and trust in the advice provided by intelligent systems. According to cognitive learning theory, an explanation is necessary for imitating behavior, which happens in consultation with human experts (Gregor & Benbasat, 1999). By providing adequate justifications for the advice systems offered, explanations can lead to greater trust in the whole intelligent system (Gregor & Benbasat, 1999). An early study in recommendation systems by W. Wang and Benbasat (2007) found that transparency can improve human trust in the systems. Wang et al. (2016a) designed a human-robot interaction experiment and found that explanation improves transparency and thus improves human trust in the decision given by robots. Similarly, Hussein et al. (2020) found that transparency and reliability are the critical elements for trust-building in human-swarm interaction. Trust indicates high reliance rates and shorter response times. Thus, in this study, *we propose that in competitive commercial prediction tasks, explainable AI improves human trust toward AI (H2)*. To test this hypothesis, three groups of participants were recruited and assigned randomly to conditions with AI decision assistance with explanation or without explanation. Both self-reported trust via a questionnaire survey and

¹ General data protection regulation (GDPR), <https://gdpr-info.eu/recitals/no-71/>.

behavioural trust of decision consistency with AI (refer to Section 3.5 for measurement details) were recorded in a control lab experiment.

However, inconsistent findings on the impact of explanation on trust in AI were found. For instance, from an *algorithm aversion* perspective, human's trust in algorithms fails after experiencing algorithm mistakes. Some research found that participants' initial trust in virtual AI decreases after the interaction; that is, they trust the AI advice more than that from a human at the beginning, and after the interaction, the trust decreases due to a lack of calibration between the high-level expectation of the machine intelligence and the actual level of intelligence (Hoff & Bashir, 2015). From an *automation bias or algorithm appreciation* perspective, people tend to overly trust automation or algorithms (Cummings, 2004).

According to the elaboration likelihood model (ELM), individuals may be persuaded via dual routes, the central core routine of argument quality or the peripheral routine by information cues (Petty & Cacioppo, 1986). For the *central persuasion route*, individuals appeal for a rational and quality argument to be persuaded. In AI-assisted decision-making, the central persuasion routes refer to the rational explanations of how the AI prediction result is derived (Burton et al., 2020). Contrarily, individuals refer to heuristic cues for decision-making rather than thoughtful information content processing for the peripheral route. People with different elaboration likelihoods would process information in different ways. Users with less data analysis capacity are defined as a *low task-related capacity group* in the current study and tend to be persuaded via peripheral cues without intensive information content. Whilst, experts with high data analysis capacity, referred to as a *high task-related capacity group* in this study, tend to be persuaded via the central routines of argument quality. The details of the measurements for *low/high task-related capacity* will be presented in Section 3.5. In the case of explainable AI, with different task-related capacities, individuals might be persuaded via the explanation information content (argument quality) or the existence of the explanation (cue), contributing to the decision performance and trust-building toward AI. Therefore, it is worthy of further exploration of this question by distinguishing the impact of the rationality and cue of explainable AI. Thus, we propose that:

H3a: The rationality of explanation is essential in improving prediction performance for high-capacity people, while the cue is enough for the low-capacity group.

H3b: The rationality of explanation is essential in improving trust in AI for high-capacity people, while the cue is enough for the low-capacity group.

Participants in our experiments will be divided into low and high data analysis capacity groups based on the data analysis test, decision performance of their prediction accuracy, average time cost for completing the tasks, self-report trust and behavioral trust will be compared to test H3a and H3b.

3. Methodology

3.1. Study design

To test the hypotheses mentioned above, we conducted a laboratory experiment with our own developed experiment platform. Using a quasi-experiment (nonequivalent control group) design, three groups completed the experimental task. All groups make decisions with only AI predictions in the first ten trials. In the second ten trials, different explanation conditions were given. In this study, feature contribution explanations generated by a state-of-the-art explainable algorithm, SHAP (Shapley Additive exPlanations, denoted as XAI in the current study), were adopted (Lundberg & Lee, 2017). SHAP can offer robust and reliable explanations for “black-box” outcomes independent of the type of algorithm model used and provides mathematical guarantees for the accuracy and consistency of explanations (Lundberg & Lee, 2017). It is an established explanation technique for complex AI models and one of the most widely used model-agnostic explanation tools, which assigns each feature an importance value for contributing to a particular algorithm prediction decision. Feature contribution explanation based on SHAP is not sensitive to perturbations in the data compared to other model-agnostic explanation technologies (e.g., LIME, *Local Interpretable Model-Agnostic Explanations*) (Ribeiro et al., 2016). SHAP explanation method has been evidenced to be effective in improving people's trust in complex AI models and empowering people to trust AI appropriately (X. Wang & Yin, 2022). The explanation was visualized with SHAP's waterfall plots (as presented in Fig. 3–3 below.)”

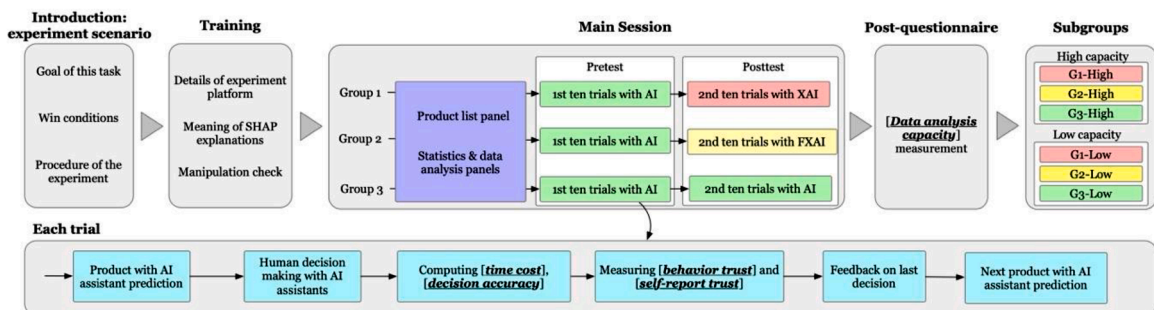


Fig. 3–1. The framework of our study design.

Participants in the control group make decisions with *AI Assistant I* which provides only prediction results without explanation (denoted as AI, Group 3). Another group makes decisions with *AI Assistant II* (Group 1) which provides both prediction results and rational explanations generated by SHAP. The third group make decisions with *AI Assistant III* which provides prediction results and randomly generated explanations (denoted as FXAI, Group 2). Fig. 3–1 shows the framework of our study design and the overview of our experimental procedure.

Firstly, a comprehensive introduction lecture was delivered to participants, aiming to establish a clear understanding of the task's objective, the task scenario, and the criteria for winning the task. Please see details in Section 3.2.

Secondly, an interactive training session was implemented to familiarize participants with the experimental platform usage and the interpretation of SHAP explanations. In the training, the meaning of the SHAP explanation was explained in detail to the participants, including the different colors of the bars in SHAP, and the attributes that appeared in the explanation. The explanation materials presented in the training session are presented in Appendix B. To ensure participants' accurate understanding of SHAP plots, a manipulation check question (refer to Appendix B) was posed at the end of this session. Participants will continue reviewing the training material until they successfully pass the check. Details are presented in Appendix B.

Thirdly, the decision-making session was scheduled to gather data from participants. Each participant was required to perform 20 prediction trials with the help of various AI assistants. During this session, factors such as time taken, decision accuracy, behavior trust, and self-report trust were recorded and measured. Further information can be found in Section 3.3.

Fourthly, we employed a post-questionnaire to assess the participants' capacity in data analysis. This enabled us to categorize them into subgroups for subsequent analysis. Details are presented in Appendix A.

3.2. Task scenario

We set up an e-Commerce Sales Forecasting (in the following, referred to as ECSF) task. Participants must forecast sales and make decisions on some candidate products in collaboration with AI assistants. Participants in our ECSF task are assigned the role of sellers in a cross-border e-commerce platform; they should first browse a list of selling products and analyze the relationships among product attributes and sales performance, and then make predicting decisions on some candidate products with the help of different types of AI assistants. Specifically, in the ECSF task, candidate products are shown one by one, and participants are required to make predicting decisions on whether they think the sales of each candidate product will be higher than 10,000 pieces or not.

To simulate a competitive commercial scenario, three levels of monetary bonus were linked to the decision performance of participants. Participants were ranked by prediction accuracy first; if participants had the same prediction accuracy, then the one who took a shorter time wins. The top 5 % of participants can get a reward of 100 RMB, the top 5 %–15 % of participants win 80 RMB, and the rest of them receive 50 RMB as a reward for participation. The setting of the monetary bonus based on their performance compared to peers makes them have a competitive consciousness and motivates the participants to be cautious in the decision outcome (Zhang et al., 2020).

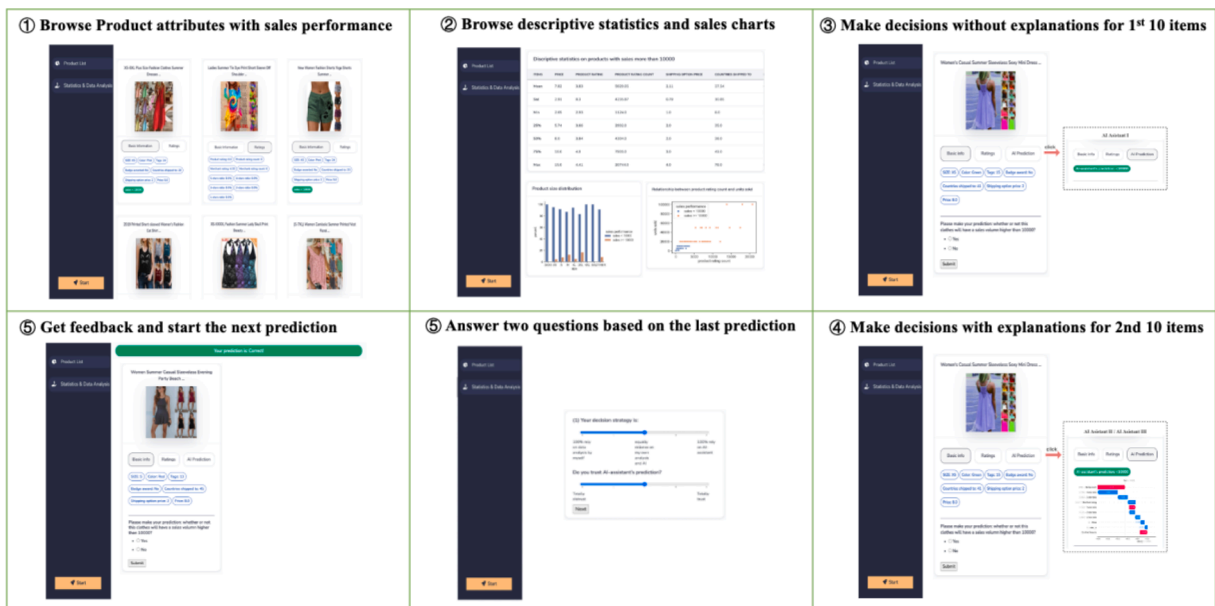


Fig. 3–2. Decision-making procedures.

3.3. Decision procedures, platform development

3.3.1. Procedures in decision-making session

Three hours time limit was set for the experiment with 20 prediction trials so that participants could have enough time to browse selling products in the product panel, analyze tables and graphs in the statistics and data analysis panel, and make predictions for decision items (i.e., the set of candidate products).

Fig. 3–2 shows the procedures on our experiment platform. Upon participants log in to the experiment system, they can browse a list of selling products with basic information and user-generated content, such as price, color, size, shipping option price, badge awarded, product rating, merchant rating, etc. The sales performance of each selling product has also been highlighted in colors (i.e., Red for sales $\geq 10,000$ and green for sales $< 10,000$). Besides, statistical tables, charts, and graphs of products are provided further to analyze the associations between product attributes and sales volume. Participants can start the prediction task anytime when they are ready and can revisit the product panel and the data analysis panel during prediction tasks. Participants are asked to finish 20 prediction tasks. For participants in all three groups, in the first ten trials with AI prediction results, participants need to make decisions with AI Assistant I, i.e., hints of sales performance for each decision item without explaining why the AI assistant makes the prediction. In the second 10 trials, participants in Group 1 and Group 2 are offered the prediction results and explanations (i.e., either AI Assistant II or AI Assistant III) describing how each feature contributes to the prediction result, and participants in Group 3 still make decisions with AI Assistant I without any explanations. After each decision, two questions about their trust perception and reliance on AI were asked (the time cost on these two survey questions is excluded from the total time cost of this task). Then participants were shown feedback on whether their decision was correct for the last prediction.

3.3.2. Platform development

To conduct the ECSF task, we developed an experimental platform consisting of three key components: (i) a dataset collected from



Fig. 3–3. Screenshot of different AI assistants.

the real cross-border e-commerce platform; (ii) three different types of sales performance prediction AI assistants; (iii) a web user interface.

3.3.2.1. Experimental dataset. To make our experimental task more realistic, we collected 1537 products from a publicly available dataset on Kaggle (refer to [Appendix C](#) for the details of the data utilized). All products will appear on Wish.com (i.e., a real cross-border e-commerce platform) if you type “summer clothes” in the search bar in early August 2020. Each product is associated with features including basic product information (e.g., title, price, size, color, etc.), user-generated content (e.g., product rating, number of tags, merchant rating, etc.), and sales volume. We randomly split all products into two parts: a learning dataset (50 %) and a predicting dataset (50 %). The learning dataset is utilized for browsing, data analysis, and training sales prediction model (i.e., one of the core parts of AI assistants). All products in the predicting dataset are used to test the performance of the trained sales prediction model, and 20 of them are selected for sales volume prediction for our experimental task.

3.3.2.2. AI assistants. To answer the two research questions proposed in the current study, we developed three different AI assistants by combining a popular machine learning algorithm and an explainable artificial intelligence technique. Firstly, we trained a sales prediction model on the learning dataset using the XGBoost algorithm ([Chen & Guestrin, 2016](#)); the prediction model can classify each candidate product into two categories: “sales $\geq$ 10,000” and “sales $<$ 10,000”. The trained prediction model is utilized for developing a sales prediction assistant without explanations in our experiment platform, which we refer to as *AI assistant I* in this paper (see [Fig. 3–1\(a\)](#)).

Secondly, we applied SHAP, a model-agnostic explainable AI technique proposed by [Lundberg et al. \(2020\)](#), to generate explanations for the prediction model. The SHAP explanation plot shows how each feature contributes to changing the prediction model’s output. Features pushing the sales volume prediction higher are shown in red, and those pushing the prediction lower are in blue. For example, the SHAP explanation plot in [Fig. 3–1\(b\)](#) shows that the 55.8 % of 5-star rating gives a 0.32 positive influence on the prediction of sales volume, and the price of 8 EUR gives a 0.22 negative effect. AI prediction results are integrated with explanation plots generated by SHAP for developing *AI assistant II*, i.e., a sales prediction assistant with rational explanations (see [Fig. 3–1\(b\)](#)). In our experiment setting, the ‘rational explanation’ refers to the explanations plotting the actual contribution of each product attribute generated by SHAP. The contribution of product attributes on sales volume is consistent between products compared to the irrational explanations for *AI assistant III* in the following.

Third, we modified *AI Assistant II* to create *AI Assistant III* by adding a randomness algorithm into the SHAP explanation plot generation process, i.e., a sales prediction assistant with ‘irrational’ explanations (see [Fig. 3–1\(c\)](#)). It shares the same sales prediction model and similar plot appearance as *AI assistant II*, but contribution values were randomly permuted among all features. For instance, in [Fig. 3–1\(c\)](#), 55.8 % of 5-star rating gives a 0.4 negative influence on the prediction of sales volume, but 20.6 % of 4-star rating has a 1.4 positive influence. Thus, the contribution of high ratings (5-star and 4-star) to sales volume is contradicted, which forms the so-called irrational explanations generated by adding the randomness algorithm into SHAP for Assistant III. To guarantee all participants understand the explanation generated by SHAP, we introduced the plots as depicted in [Fig. 3–3](#) to all participants before they started the prediction tasks. As a prerequisite, to make sure all participants understand SHP plots and how each feature contributes to the final AI prediction results, participants need to identify the most critical features that positively and negatively contribute to the prediction results. They can continue to start after they answer it correctly.

3.3.2.3. Web interface. The web user interface of our experiment platform consists of three main panels: (i) a product panel, where participants can browse a list of selling products with basic information and user-generated content; (ii) a statistics and data analysis panel, where statistical tables, charts, and graphs for two categories of products (i.e., “sales $\geq$ 10,000” and “sales $<$ 10,000”) are provided to participants for further analysis; (iii) a decision panel, where the small set of sampled products are displayed one by one and participants can click the button to submit their decisions with the help of different types of AI assistants. After the submission, two survey questions about their trust perception and reliance on AI were asked (refer to Step 5 in [Fig. 3–2](#) for details). Participants are free to switch among different panels by clicking the navigation tabs on the left of our experiment platform.

Table 3–1
Method for hypotheses testing.

Hypotheses	Groups comparison	Variables to be compared
H1a	G1 – G3 G2 – G3	accuracy
H1b		time
H2		trust, reliance and consistency
H3a	G1(Low) - G3(Low)	accuracy and time
H3b	G2(Low) - G3(Low)	trust, reliance, and consistency
	G1(High) - G3(High) G2(High) - G3(High)	

Note: “Low” represents participants of low task-related capacity, and “High” represents those of high task-related capacity.

3.4. Participants

According to the nomogram by Machin et al. (2018), in two groups comparison, for an effect size of 1, to achieve a statistical power of 0.8 at a significance level of 0.05, the sample size should be at 30. Thus, in our experiment, we recruited 124 participants altogether, and the final valid sample size is 36 for Group 1, 33 for Group 2, and 40 for Group 3. The participants are selected according to the following criteria and reasons. First, all selected participants are majoring in e-commerce or a related major such as information systems management and information resource management. Thus, all participants have a basic knowledge of e-commerce and have already finished a data analytics course suitable for our ECSF task. Secondly, students without e-commerce working experience were preferred over industrial experts, as industrial experts on e-commerce would have more consideration for their subjective knowledge and prior experience. The subjective understanding of human decision-making of a specific task will contribute to the algorithm aversion (Cadario et al., 2021), which would be a threat to the internal validity of our experiment.

For the first two groups, participants were assigned randomly to Group 1 and Group 2, as depicted in Table 3–1. To improve the internal validity of our experiment, a third group of participants were recruited. After the experiment, we screened data for survey questions asking about their trust and reliance on AI. Those with all answers the same were excluded. The final valid participants are 36 in Group 1, 33 in Group 2, and 40 in Group 3. Participants in all three groups are required to make decisions with AI Assistant I for the first ten products (Step 3 in Fig. 3–2), then AI Assistant II is provided to participants in Group 1 and help them make decisions on the other ten products (Step 4 in Fig. 3–2). Still, participants in Group 2 need to make decisions with the help of AI Assistant III for the last ten products. Group 3 is the control group with only AI for assistant, thus, participants in Group 3 are required to make all the 20 prediction tasks with Assistant I.

3.5. Measurements

3.5.1. Self-report trust

In self-report trust, *trust* perception, and *reliance* on AI were measured by asking two questions. Participants were asked the question “Do you trust the prediction justification offered by AI decision assistants?” and “Your decision strategy?” whenever they made a decision. Participants can move the pointer on a horizontal cursor, ranging from “I totally trust AI prediction” (trust score=100) to “I don’t trust the AI prediction at all” (trust score=0) for the first question, and “100 % rely on AI hints” (reliance on AI=100) to “100 % rely on data analysis by myself” (reliance on AI=0).

3.5.2. Behavioral trust

The behavioral trust was measured by calculating the answer *consistency* with AI, i.e., the percentage of trials in which the participant’s final prediction agreed with the AI’s prediction.

3.5.3. Performance

Performance includes the *time cost* and prediction *accuracy* for the 20 prediction tasks included in our experiments. Specifically, in our experiment platform, the timer started when participants clicked the ‘Start’ icon, and the time used for each trial was recorded. Prediction accuracy is the percentage of the right prediction trials among different AI assistants respectively.

3.5.4. High/low task-related capacity

To distinguish the data analysis capacity among participants, we conducted a post-experiment survey, and 7 “True or False”



Fig. 3–4. A screenshot of the sales plot.

questions were asked based on their understanding of the statistical description of the sales from our experiment platform (refer to the screenshot in Fig. 3–2②), which depicted the relationships between sales performance and product attributes (refer to Appendix A for quiz details). All the questions are designed to test participants' data analysis capacity. From the plots depicting the relationships between product attributes and sales performance, some rules were included. For instance, there are no clothes with sales of more than 10,000 pieces if the size is "3XXS" nor a price higher than 15\$, as depicted in Fig. 3–4. Participants' data analysis capability can be differentiated based on whether they can identify the rules successfully. The reason for a post-experiment test rather than a pre-experiment test for the capability is to avoid the contextual cueing effect, i.e., reminding or guiding the participants to analyze data via the way included in the True/False quiz question. The participants were divided into two groups: the *high task-related capacity* with a True or False quiz score between 4 and 7 and the *low task-related capacity* with a score between 0 and 3.

To test Hypotheses 1a, 1b, and 2, participants' decision accuracy, average time cost for each prediction task, consistency with AI prediction, self-reported trust for the 2nd 10 trials between Group 1 and Group3, Group 2 and Group 3 were compared (details in Table 3–1). To test Hypotheses 3a and 3b, participants were divided into high or low task-related capacity groups according to their performance in the *post-experiment True or False* test (refer to Section 4.3 and Appendix A for details).

Before the data analysis for checking the effect of treatments, the difference in their trust in AI, reliance on AI, average time cost for the 1st ten trials, accuracy for the first ten trials, between Group 3 and Group 1, Group 3 and Group 1 before treatments were tested (Campbell & Stanley, 2015; Cook et al., 2002). No significant differences were found, and the details of the test are presented in Table 3–2.

4. Results

To answer the two primary research questions and test the five hypotheses proposed above, we conducted paired sample *t*-test based on the data collected via experiments. The results from pared sample *t*-tests are presented in Table 4–1; the effect sizes are reported with Cohen's *d* values at the 95 % confidence interval of the difference. According to the effect size results, we can find that offering rational explanation (i.e., Assistant II) exerts medium effects on the prediction accuracy (Cohen's $d = 0.737$), reliance on AI (Cohen's $d = 0.747$), and human consistency (Cohen's $d = 0.627$) with AI. Specifically, the average prediction accuracy in the condition with Assistant II is 0.737 standard deviations higher, reliance on AI prediction is 0.747 standard deviations higher, and consistency with AI 0.627 standard deviations higher, respectively, than that with AI.

Presenting irrational explanation (Assistant III) exerts a relatively small effect size for human reliance on AI (Cohen's $d = 0.494$), and medium effect size on prediction accuracy (Cohen's $d = 0.624$).

In the following section, we will present the results of the three main research questions and five research hypotheses proposed at the beginning of our research.

4.1. Explainable ai and human performance

The first research question is whether the presence of explanation can improve human performance, either in decision accuracy or decision time. Previous studies have inconsistent findings on the addition of algorithm transparency and human performance. We compared the 2nd ten trials of participants' average time cost and prediction accuracy in Group 1 and Group 3, and Group 2 and Group 3 by paired-sample *T*-test. We found that the addition of an explanation cannot improve the decision efficiency significantly in both groups, neither for Group 1 with a rational explanation ($\Delta_{\text{time}} = -0.527$ s, $p = 0.785$, *n.s.*) nor for Group 2 with a randomly generated explanation ($\Delta_{\text{time}} = 3.786$ s, $p = 0.145$, *n.s.*), as shown in Fig. 4–1(a). Thus, *Hypothesis 1a* was not supported. However, the prediction accuracy was improved significantly with the presence of explanation, with an increased accuracy ratio of 0.108 ($p < 0.01$) for Group 1, and an increase of 0.092 ($p < 0.01$) for Group 2, as depicted in Fig. 4–2(b). Thus, *Hypotheses 1b* was supported.

4.2. Explainable ai and human trust in ai

In prior research, both self-reported trust and behavioral trust indicators were employed to measure human trust in technology, and inconsistent findings were reported in the abovementioned discussion. To test whether or not the presence of explainable AI improves human trust in AI, we employed both self-report trust and objective behavioral trust measurements.

Our results indicate that XAI cannot improve participants' self-report trust perception toward AI significantly, neither rational explanation in Group 1 nor randomly generated irrational explanation in Group 2. Fig. 4–2(a) shows the mean trust in AI among the two groups. However, the reliance on AI improved significantly with the addition of explanation. As depicted in Fig. 4–2(b), reliance

Table 3–2

Differences in trust, reliance, time, accuracy, and consistency for the 1st ten trials.

	Time	Accuracy	Reliance	Trust	Consistency
G1(AI) – G3(AI)	1.786[0.681]	0.020[0.556]	4.028[0.175]	–2.394[0.237]	0.022[0.494]
G2(AI) – G3(AI)	5.729[0.280]	0.006[0.863]	–1.506[0.660]	–0.505[0.832]	0.008[0.924]

Note: Values in square brackets indicate the *p* values of the differences.

Table 4-1Results from pared sample *t*-test for the 2nd 10 trials between groups.

Pairs on 2nd ten trails		Performance		Self-report trust		Behavioral trust consistency
		Time	Accuracy	Reliance	Trust	
Group1 XAI (Assistant II) – Group 3 AI (Assistant I)	Paired differences	–0.527	0.108**	9.385**	3.319	0.084**
		[–4.43, 3.37]	[0.04, 0.17]	[3.61, 15.16]	[–1.54, 8.18]	[0.02, 0.14]
	Effect sizes	0.062	0.737	0.747	0.317	0.627
Group 2 FXAI (Assistant III) – Group 3 AI (Assistant I)	Paired differences	3.786	0.092**	6.220*	2.265	0.051
		[–1.67, 9.24]	[0.02, 0.16]	[0.25, 12.19]	[–2.65, 7.18]	[–0.02, 0.12]
	Effect sizes	0.346	0.624	0.494	0.221	0.343

Note. Values in square brackets indicate the 95 % confidence intervals. Effect sizes are reported with Cohen's *d* values at the 95 % confidence interval of the difference. Asterisks indicates the significance level: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

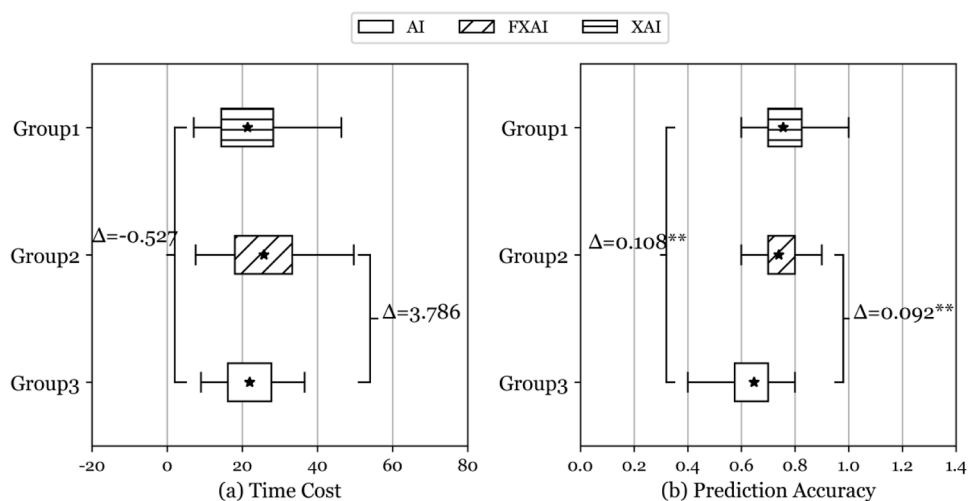


Fig. 4-1. Boxplots for participants' time cost (a) and prediction accuracy (b) in each condition. Notes. Star in the box indicates the average value. AI=prediction with Assistant I, XAI=prediction tasks with Assistant II, FXAI=prediction tasks with Assistant III (irrational).

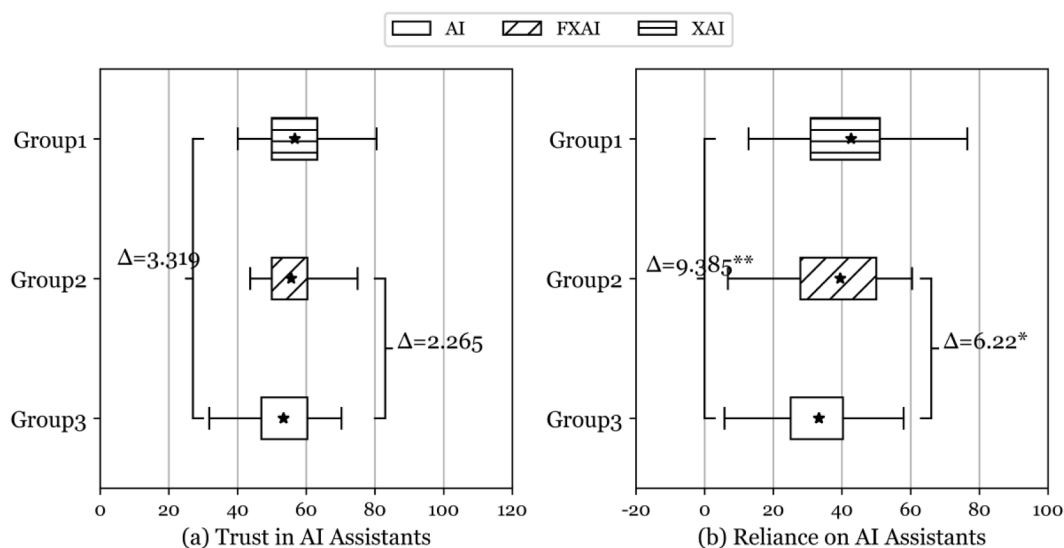


Fig. 4-2. Boxplots for participants' self-reported trust (a) and reliance (b) on AI in each condition. Notes. Star in the box indicates the average value. AI=prediction with Assistant I, XAI=prediction tasks with Assistant II, FXAI=prediction tasks with Assistant III (irrational).

on AI increased by 9.385 ($p < 0.01$) for Group 1 and 6.22 ($p < 0.05$) for Group 2.

We compared participants' mean consistency in the experiments across conditions using the paired-sample T-test (see Fig. 4–3). We found that for Group1, where the rational explanation for the AI prediction results was presented, the consistency increased significantly by 8.4 % ($p < 0.05$). The increase of consistency with AI results was not significant for Group 2, where the explanations were generated randomly. Thus, **Hypothesis 2** was partly supported.

4.3. The task-related capacity and explanation

With inconsistent findings on the impact of transparency, precisely explanation, on human trust in AI, this study intended to explore the influencing routines of explanation among participants and the role of the rationality of explanation. According to ELM theory, people with different task-related capacities, i.e., data analysis capacity, might be persuaded via different routes, the core routine of informal rationality, or the peripheral cue such as the presence of transparency.

We compared the average time cost for the task, reliance on AI, self-report trust in AI, and the accuracy of prediction and answer consistency with AI. Table 4–2 shows that, for low task-related capacity (data analysis capacity in this research), the cue of explanation is enough to improve human decision accuracy, and the reliance on AI. Specifically, for participants with low task-related capacity, as depicted in Table 4–2, no matter with rational or irrational explanations, their prediction accuracy increased significantly in both Group 1 ($\Delta_{\text{Accuracy}}=0.111$, $p < 0.01$) and Group 2 ($\Delta_{\text{Accuracy}}=0.094$, $p < 0.05$). For low task related capacity, the existence of an explanation interface is enough to enhance their reliance on AI ($\Delta_{\text{XAI-reliance}}=9.746$, $p < 0.01$, $\Delta_{\text{FXAI-reliance}}=8.51$, $p < 0.05$). Thus, Hypothesis 3a is partly supported.

For participants with relatively high task-related capacity, prediction consistency with AI increased significantly with the presence of explanation, and the increase in rational explanations is higher than in irrational cases ($\Delta_{\text{XAI-consistency}}=0.268^{**}$, $p < 0.01$, $\Delta_{\text{FXAI-consistency}}=0.205$, $p < 0.01$), which is different in the condition for low-capacity people. This indicate that, explanation plays a more important role for those with relatively high data analysis capacity.

To further unraveling the different impact routines of explanations for people with different capacity level, we went through the details of participants behavioral consistency with AI. The accuracy of AI prediction model is 75 % in our research, thus, among 20 prediction tasks, there were five mistakes by AI prediction altogether. While completing the prediction task, participants would have the experience that AI makes more and more mistakes. Thus, for those participants with high capacity, they would decide to depend on themselves, thus their behavior trust, i.e., consistency with AI, decrease dramatically during the 2nd ten trials in AI condition ($\Delta_{\text{Group3,consistency}}=-0.175^{**}$, $p = 0.007$). Thus, when presented explanations for AI prediction results to high-capacity participants, their behavioral consistency with AI prediction increases significantly. Comparatively, low task-related capacity people tend to be more consistency with AI, no matter with explanations or not, regardless of the rational of explanations.

In conclusion, the presence of explanation is essential in improving the behavioral trust for high-capacity people, but not in improving their self-report trust. Hypothesis 3b is partly supported.

The summary of the hypotheses testing results is presented in Table 4–4. The results of our study supported Hypothesis 1a confirming that adding an explainable interface can improve AI-assisted human decision accuracy, but declined Hypothesis 1b, i.e., cannot improve decision efficiency. Explainable AI can improve reliance on AI, and rational explanation can improve behavior trust on AI but not the self-report subjective trust to AI. Thus, Hypothesis 2 is partly supported. For low task-related capacity people, the existence cue

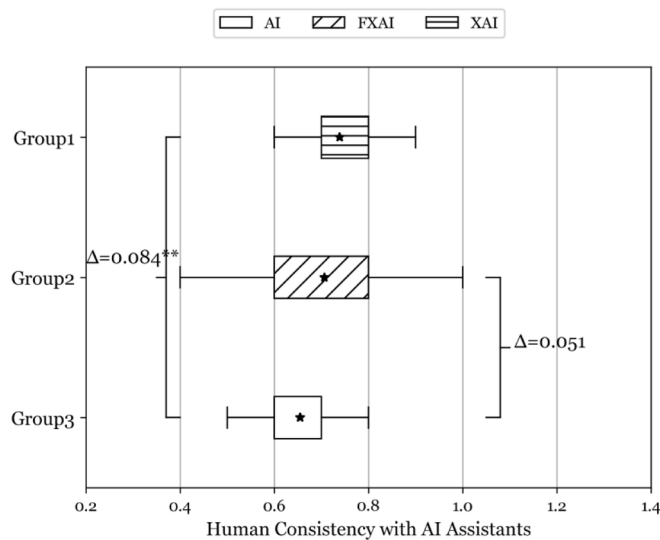


Fig. 4–3. Boxplots for participants' average prediction consistency with AI in each condition. Notes. Star in the box indicates the average value. AI=prediction with Assistant I, XAI=prediction tasks with Assistant II, FXAI=prediction tasks with Assistant III (irrational).

Table 4-2

Delta time, reliance on AI, trust in AI prediction, right ratio, the consistent ratio with AI prediction for low and high task-related capacity participants.

Task-related Capacity	Group pairs	Decision performance		Self-report trust		Behavioral trust
		Time	Accuracy	Reliance	Trust AI	Consistency
Low	XAI-AI	-0.384	0.111**	9.746**	4.454	0.063
Low	FXAI-AI	2.982	0.094*	7.244*	2.061	0.053
High	XAI-AI	-0.497	0.082	11.264	1.889	0.268**
High	FXAI-AI	5.754	0.065	10.403	7.768	0.205**

Notes. $p < 0.05^*$; $p < 0.01^{**}$; $p < 0.001^{***}$; in group pairs, all AI conditions are the 2nd 10 prediction in Group 3, XAI are the 2nd 10 prediction with rational explanation in Group 1, and FXAI are the 2nd 10 prediction with irrational explanation in Group 2.

of explanation is enough to improve prediction accuracy, and their reliance in AI, but not prediction time nor behavioral trust, thus Hypothesis 3a partly supported. Explanation is essential for high task-related capacity people in improving their behavioral trust to AI, but not in prediction performance nor self-report trust, thus, Hypothesis 3b partly supported.

5. Discussion

This research is motivated by the query of whether or not transparency of presenting explanations can improve human decision performance and trust in AI in the competitive decision-making tasks and how the explanation impact individuals with different task-related capacity. Correspondingly, in our experiment setting, a competitive commercial decision-making scenario was selected to motivate participants to process the explanation information critically to unravel the first research objective. To investigate the impact of routines of explanations for individuals with different capacities, we divide the participants into two groups according to their post-experiment true/false quiz scores.

5.1. Impact of xai on performance and trust in competitive commercial tasks

5.1.1. XAI improves AI-assisted human decision accuracy in competitive commercial prediction tasks

The results indicate that with explanation, AI-assisted decision-making accuracy can be improved significantly in the competitive commercial prediction tasks. This finding contrasts with prior experiment studies, especially by Schmidt et al. (2020) and (Zhang et al., 2020). In the behavior experiment of text classification by Schmidt et al. (2020), participants could receive an additional reward if they achieved a classification accuracy of 85 % or above. Participants were informed that model accuracy was not perfect. The feedback was not presented to the participant to avoid learning. The setting of making ML's prediction accuracy (80 %) was controlled to be lower than the objective accuracy (85 %) for participants. The lack of feedback might be the reason for finding a negative explanation impact on human prediction accuracy. The insignificant impact of explanations on human-AI trust was also proved by Zhang et al. (2020)'s experimental study of income prediction. In most extant experimental exploration on this, the task objective is isolated with an aiming accuracy. To make participants process explanation information more critically and thoroughly, we informed the participants of the competitive scenario. Feedback on their prediction accuracy was also provided timely after their prediction to calibrate trust in AI and have an appropriate perception of AI prediction accuracy.

In the AI-assisted decision-making scenario, humans are the final agent in charge of decision-making. To test whether AI-assisted decision accuracy exceeds that of AI alone, we compared the decision accuracy of AI alone, AI with humans, and XAI with humans. The results in Table 5-1 show that, with AI prediction hints, human decision-making accuracy in both groups is lower than that of AI prediction accuracy (0.75). While, given rational explanations for AI prediction results, the human decision accuracy increased significantly and higher than that of AI alone and achieved the so-called *hybrid intelligence* (Dellermann et al., 2019).

5.1.2. XAI didn't improve human-AI self-report trust in competitive prediction tasks

The results didn't support our Hypothesis 2 that explanation could improve human self-report trust in AI but improved their reliance on AI. This finding is, to some extent, confirms the state that self-report trust is not a reliable indicator for trusting behaviors in AI-assisted decision tasks (Zhang et al., 2020). However, in our results, the prediction accuracy for the two groups increased significantly with explanations presented, but the behavioral trust toward AI only increased significantly in Group 1 with rational explanations. This indicates that people are calibrating their trust behaviors across the prediction tasks. Though the AI-assisted decision performance also increased in Group 2 with irrational explanations, the participants' behavioral trust (consistency with AI) didn't improve significantly with explanations (as shown in Table 4-1). It is stated that offering local explanations can contribute to trust calibration in AI (Lai & Tan, 2019). Trust calibration is the process of building appropriate trust in AI, neither overly trust nor aversion to algorithms. In our experiment setting, the feedback on each prediction task was presented so that participants could calibrate their trust in AI prediction and moderate their prediction strategy, either by themselves, totally trust AI, or use AI as a complementary cure for human errors. Our findings by taking the rationality of explanation in our experiment setting offer evidence that showing local rational explanations can help people build appropriate trust in AI.

5.2. Impact of XAI on performance and trust for different task-related capacity

5.2.1. XAI improves AI-assisted decision-making accuracy and reliance on AI for low task-related capacity people, regardless of the explanation rationality

By considering explanation rationality, we find that the cue of explanation can improve human decision accuracy, and reliance on AI for low task-related capacity. Our finding is consistent with the tenet of the elaboration likelihood persuasion model by Petty and Cacioppo (1986). People with less motivation or lack the necessary ability for data analysis would process information via a peripheral route and make a heuristic decision. In AI-augmented decision-making, scholars found it necessary to classify decision-makers as heuristic and analytic (Burton et al., 2020). In AI-assisted decision-making, people with less expertise in data analysis would be classified as heuristic decision-makers. They rely on information cues without critical or careful thinking of the information content. Thus, the presence of an explanation interface is sufficient to improve their decision accuracy. Their prediction consistency with AI remains at a similar level with or without explanation, which further confirms that low task-related capacity people would adopt AI results regardless of the rationality of the explanation (as indicated in Table 4–3). Prior findings also found that decision-making for experienced people concentrates more on analytical description information while less experienced ones rely more on the information cues arousing the attention (Lin & Boh, 2021). This finding evidenced the *placebo effect* of explanation in the human-AI interaction during the AI-assisted tasks for a specific group of people, which is also demonstrated in previous HCI studies (Kosch et al., 2022). Specifically, during human-AI interaction, participants might have positive expectations of systems' performance before their interaction, especially for novel technology, thus leading to the placebo effect in human-computer interaction, such as the fake power-up elements in gameplay (Denisova & Cook, 2019). Thus, in our study, for low task-related capacity people, though the irrational explanations didn't actually improve the transparency of the algorithm nor help to understand the AI prediction results, they still have better decision accuracy and reported higher reliance on AI in the condition of irrational explanations.

6. Conclusion

XAI is an increasingly important research focus from various disciplines. With the motivation to harness high-performance AI models in “black-box,” research quest explainability as a potential measure to improve the trust in AI and the adoption of AI algorithms. Instead of being persuaded to accept AI results or predictions, it is more critical to build appropriate trust in AI, that is, and to achieve better performance with AI assistants. In this study, we have investigated whether explainable artificial intelligence can improve the trust in the AI results and the AI-assisted decision performance in competitive tasks and take the rationality of explanation and human factor of task-related capacity into an experiment setting. Overall, we confirmed that improving transparency by offering explanations can improve AI-assisted human decision-making accuracy in competitive commercial prediction tasks. By comparing human factors of task-related capacity, we found the placebo effect of the AI explanations for low task-related participants. People with different task-related capacity differs in their trust calibrating process during the interaction with AI.

6.1. Theoretical implications

This study contributes to innovative technology usage research from an interdisciplinary perspective by uncovering the trust in human-AI interaction. Increasing algorithm transparency by explaining how the algorithm works are believed to be an excellent way to increase trust in AI (Wang et al., 2016b). Thus, many scholars have called for transparency for intelligent systems to improve trust in IT artifacts, especially in the intelligent systems (Glikson & Woolley, 2020). Though prior studies have explored trust in IT artifacts, trust is primarily measured by self-report trust via survey questions. Our study findings attest that explanation cannot improve the participants' self-report trust in AI in complex prediction tasks with massive data analysis but can improve the behavioral trust in AI in the way of increased consistency with AI prediction results.

Secondly, this study contributes to our understanding of various persuasion routines of AI decision aids by considering the rationality of explanation. People with high task-related capacity focus more on the explanation rationality, i.e., in our explanation model, whether the main product features identified by the ML algorithm contributing to or reducing the probability of a high-sale product is reasonable. Meanwhile, with low task-related capacity, people can be persuaded via simply the cues of adding an explanation for AI prediction results without critical thinking about the content of the explanation.

This study also extended our knowledge of the influence of XAI on human performance in competitive commercial decision-making. Scholars have called for the human-algorithm ensemble, especially in the managerial decision-making (Choudhary et al., 2021). Our results evidenced that human accuracy with XAI increased significantly compared to that of AI alone and humans with AI. Thus, with an explanation of the AI prediction results, we can achieve hybrid intelligence.

Table 4–3

Average prediction consistency with AI for all three groups.

	Group1		Group2		Group3	
	Low	High	Low	High	Low	High
1st ten trials with AI	0.686	0.6	0.656	0.667	0.658	0.650
2nd ten trials with XAI (G1), FXAI (G2) or AI (G3)	0.738	0.742	0.728	0.680	0.675	0.475

Table 4–4
Summary of hypotheses test.

Hypothesis	Test result
H1a: Explainable AI improves human decision accuracy.	Supported
H1b: Explainable AI decreases human decision time.	Not Supported
H2: Explainable AI improves human trust toward AI.	Partly supported
H3a: For low task-related capacity people, the existence cue of explanation is enough to improve prediction performance and trust to AI.	Partly supported
H3b: For high task-related capacity people, the rationality of explanation is important in improving prediction performance and trust to AI.	Partly supported

Table 5–1
The average accuracy of AI alone, AI+human, XAI+human.

Metric	AI alone	AI+human	XAI+human	Delta
Group1-Accuracy	75 %	0.675 [0.3–0.9]	0.756 [0.5–1.0]	+0.081*
Group2-Accuracy	75 %	0.661 [0.4–0.9]	0.739 [0.4–1.0]	+0.078*
Group3-Accuracy	75 %	0.651 [0.3–1.0]	–	–

6.2. Practical implications

Our findings have several practical implications. First, our study shed light on our understanding of the impact of AI decision aids in managerial decision-making. Many uncertain decisions need to be made in a commercial, managerial context, especially selecting among alternatives. Decision-makers or managers need to predict with available information to achieve the best outcome. Our empirical findings indicate that, with XAI assisted in decision-making, humans can achieve the best prediction accuracy compared with AI alone and with AI assistants. This finding delivered the value of XAI application in the commercial decision-making field.

Secondly, our empirical findings identified different persuasive routines for different people. This informs practitioners to differentiate the application of AI assistants for different groups of people. For decision-makers with high task-related capability, the explanations for AI-assistant should concentrate on the information quality, especially the rationality of explanation content, as they would process the explanations critically. For decision-makers with low task-related capability, the explanations for AI-assistant can strengthen the interface design of explanation; making the cue of explanation more reliable would enhance the trust in AI-human interaction for low task-related capability.

6.3. Limitations and future research

Some limitations existed in our exploration. First, as we focused on the impact of local explanation in AI-assisted decision-making, we didn't find any significant impact on human trust in AI, however, trust is quite a subjective concept and trust building is a dynamic process. Human are prone to overtrust or distrust AI, the role of explainable AI in the dynamic trust calibrating process could be explored in future research by taking the initially estimated trust toward an AI assistant into consideration. Future research can include the process of trust calibration, and how could explanations help human build appropriate trust level to AI, and come to a stable level during the interaction. Second, our research findings indicate the placebo effect of explanation in AI-assisted decision-making. Thus, future research can further explore the impact of transparency on human trust in AI for different types of decision-makers in different task scenarios, and test our findings in field experiments with a naturalistic setting for a more fruitful view. Thirdly, only one form of SHAP explanation is offered in our experiment design, the way AI explains would also be an important factor influencing human trust, such as interactive explanation, contrastive explanation method, etc., which could be explored in the future by presenting different types of explanations in prediction tasks.

CRedit authorship contribution statement

Ping Wang: Writing – review & editing, Writing – original draft, Methodology, Funding acquisition, Data curation, Conceptualization. **Heng Ding:** Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Funding acquisition, Data curation.

Data availability

Data will be made available on request.

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Appendix A. True or False Quiz

Post-experiment test: True or False Quiz

Based on the sales prediction task and your analysis of the sales data, make your justification of the statements in the following and answer with either true or false. High sales products refer to those with a sales performance higher than 10,000 pieces.

In the clothes presented in the historical sales:

- (1) There were items of size '3XXS' with sales performance higher than 10,000 pieces.
- (2) There were items with prizes higher than 15 \$ that had sales performance higher than 10,000 pieces.
- (3) The items with more than 5000 customer reviews are all high sales products.
- (4) There is a high sales product among clothes of yellow color.
- (5) There is a high sales product among clothes of size '5XL'.
- (6) There is a high sales product among clothes of size '4XL'.
- (7) There is a high sales product among clothes with a brand rating lower than 3.5.

Appendix B. Training and Manipulation Check on the Understanding of XAI

See Fig. B1 and Fig. B2

Participants were required to complete the training before starting prediction tasks. The screenshots of the training are detailed below.

Participants have to repeatedly learn the lesson until they pass the quiz. In our experiments, 94 % of participants passed the quiz on their 1st attempt, and 100 % of participants passed the quiz on 2nd attempt. The screenshots for the feedback for those who passed the quiz (Fig. B-3) and failed the quiz (Fig. B-4) are presented below.

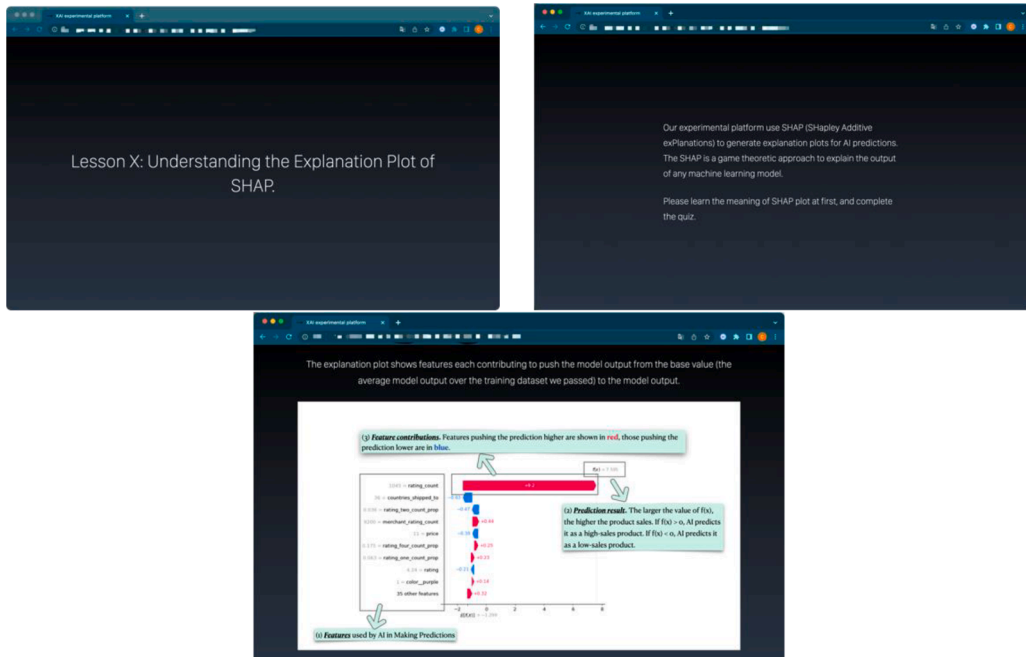


Fig. B-1. Screenshots of the materials presented in the training step.

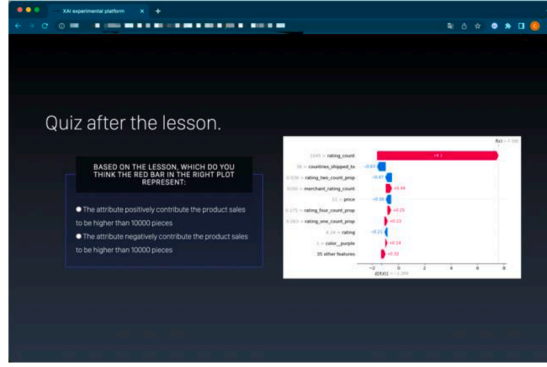


Fig. B-2. Screenshot of the quiz after training for manipulation checking.

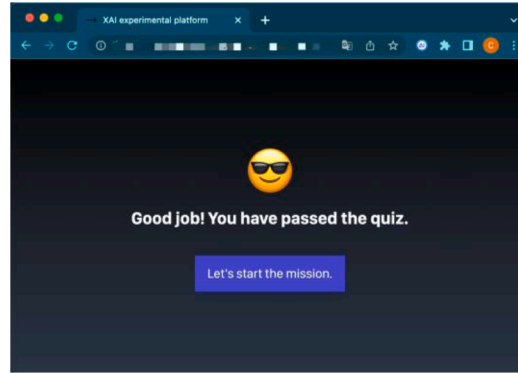


Fig. B-3. Screenshot of the feedback for those who failed the quiz.

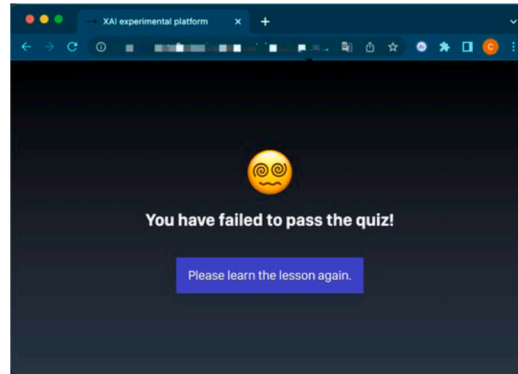


Fig. B-4. Screenshot of the feedback for those who passed the quiz.

Appendix C. Details of the data utilized in the experiment platform

We selected 1537 products from the “Sales of summer clothes in E-commerce Wish” dataset which is publicly available on Kaggle.com. The dataset contains product listings as well as various attributes regarding each product:

- title_orig: Original english title of the product.
- price: Price user would pay to get the product on wish.com
- units_sold: Number of units sold.
- uses_ad_boosts: Whether the seller paid to boost his product in wish.com.
- rating: Mean product rating based on all users who have bought the product.
- rating_count: Total number of ratings of the product.
- rating_x_count: Number of x-star ratings. X is in [1,2,3,4,5].
- badges_count: Number of badges the product or the seller have.
- countries_shipped_to: Number of countries this product is shipped to.

- `shipping_option_price`: shipping price.

More details can be found at <https://www.kaggle.com/datasets/jmmvutu/summer-products-and-sales-in-ecommerce-wish>.

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